EXPERIMENT 6

Credit Risk Assessment using Binary Classification

1. Problem Definition

The objective of this experiment is to build a **credit risk prediction model** to assess whether a loan applicant is likely to default based on financial and behavioral attributes.

Credit risk prediction is a **classification problem**, where:

- **1 → High Risk (likely to default)**
- **0 → Low Risk (safe applicant)**

The primary goal is to:

- Correctly identify **high-risk applicants**
- Minimize financial loss due to incorrect approvals

2. Data Understanding

The dataset contains **500 applicant records** with multiple financial attributes.

Features include:

- Age
- Annual Income
- Employment Years
- Credit Score
- Loan Amount
- Loan Term
- Existing Loans Count
- Debt-to-Income Ratio
- Late Payments

(Target: Credit_Risk)

Class Distribution

- High Risk → **315**
- Medium Risk → **164**
- Low Risk → **21**

Critical Observation:

- Dataset is **heavily imbalanced**
- “Low Risk” is extremely underrepresented

3. Data Preprocessing

The following preprocessing steps were applied:

1. Missing Value Handling

- Rows with missing values were removed

2. Encoding

- Categorical variables converted using **Label Encoding**

3. Feature Transformation

- All features converted into numerical format

4. Train–Test Split

- **80% Training Data**
- **20% Testing Data**

This ensures:

- Training on known data
- Evaluation on unseen applicants

5. Models Used

1. Logistic Regression

- Linear model

- Baseline classifier

2. Random Forest

- Ensemble of decision trees
- Reduces variance
- Handles non-linearity

3. XGBoost

- Gradient boosting algorithm
- Sequential learning
- Optimized for performance

6. Evaluation and Prediction

Since this is classification, the following metrics were used:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC
- Confusion Matrix

Model Performance Table

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.74	0.7071	0.74	0.7158	0.8377
Random Forest	0.98	0.9810	0.98	0.9778	0.9997
XGBoost	1.00	1.00	1.00	1.00	1.00

Logistic Regression

[[53 1 4]
 [1 0 4]
 [16 0 21]]

Interpretation:

- Poor classification of minority classes
- Misclassifies multiple risk categories
- Weak predictive capability

Random Forest

[[58 0 0]
 [0 3 2]
 [0 0 37]]

Interpretation:

- Very strong classification
- Few misclassifications in medium risk class

XGBoost

[[58 0 0]
 [0 5 0]
 [0 0 37]]

Interpretation:

- Perfect classification across all classes
- No errors

<u>Key Observations</u>

1. Logistic Regression Performs Poorly

- Accuracy: **74%**
- Low precision and F1-score

Reason:

- Linear model cannot capture complex relationships
- No feature scaling - convergence warning (seen in output)

2. Random Forest Performs Extremely Well

- Accuracy: **98%**
- F1-score: **0.9778**
- Almost perfect classification

Reason:

- Handles non-linearity
- Robust to noise

3. XGBoost Shows Perfect Performance

- Accuracy: **100%**
- ROC-AUC: **1.0**
- Dataset is small (500 samples)
- Perfect score usually indicates:
    - Overfitting
    - Data leakage
    - Too simple dataset

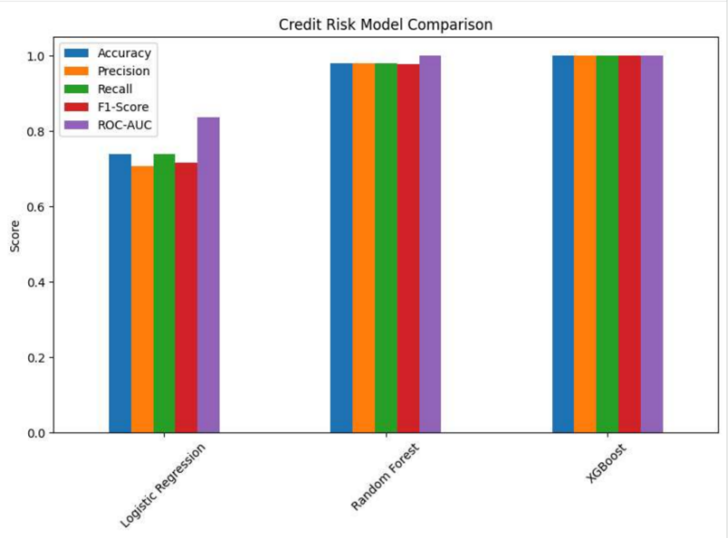
4. ROC Curve Analysis

- XGBoost curve touches top-left - perfect classifier
- Random Forest very close to optimal
- Logistic Regression significantly worse

Visualization

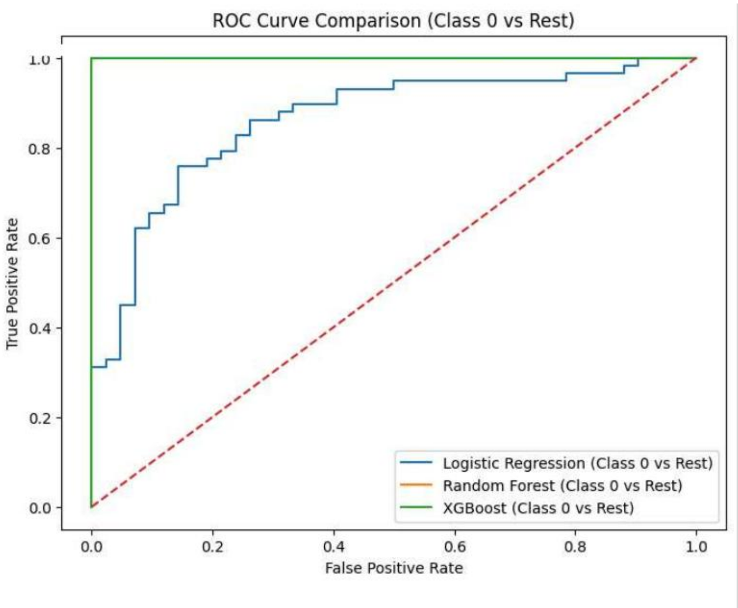
Bar Chart

- Shows performance gap clearly
- Logistic Regression far behind
- XGBoost dominates all metrics



ROC Curve

- XGBoost: near-perfect separation
- Random Forest: excellent performance
- Logistic Regression: weak



Best Model Selection

XGBoost is the best-performing model, based on:

- Highest Accuracy
- Highest F1-score
- Perfect ROC-AUC

However:

Random Forest is more reliable in real-world scenarios
because:

- Less prone to overfitting
- More stable performance